SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I P. Sai Charan Tej(192465048)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CSA07 – COMPUTER NETWORKS

CO2 – Scenario-Based Assessment 2

Solved Questions and Answers

Question 1

A Smart Campus is assigned 192.168.100.0/24. The School of Computing uses /25, the School of Electronics uses /26, and the School of Mechanical Engineering uses /27. Determine the network address, broadcast address, valid host range, usable hosts, and remaining unused range.

Solution:

- /25 has 128 addresses and 126 usable hosts. Assign the first block: 192.168.100.0–192.168.100.127.
- /26 has 64 addresses and 62 usable hosts. Assign the next block: 192.168.100.128–192.168.100.191.
- /27 has 32 addresses and 30 usable hosts. Assign the next block: 192.168.100.192–192.168.100.223.

Department	Subnet	Network	Valid Host Range	Broadcast	Usable Hosts
Computing	/25	192.168.100.0	192.168.100.1–192.168.100.126	192.168.100.127	126
Electronics	/26	192.168.100.128	192.168.100.129–192.168.100.190	192.168.100.191	62
Mechanical	/27	192.168.100.192	192.168.100.193–192.168.100.222	192.168.100.223	30

Remaining unused IP address range: 192.168.100.224–192.168.100.255.

Final Answer: Computing = 126 hosts, Electronics = 62 hosts, Mechanical = 30 hosts.

Question 2

A software company has the private IP block 10.10.0.0/16. Software Development uses /24, Quality Assurance uses /25, Human Resources uses /26, and Executive Management uses /27. Determine network/broadcast addresses, host ranges, usable hosts, and the unused range.

Solution:

- Software Development /24: allocate 10.10.0.0–10.10.0.255.
- Quality Assurance /25: allocate the next block, 10.10.1.0–10.10.1.127.
- Human Resources /26: allocate the next block, 10.10.1.128–10.10.1.191.
- Executive Management /27: allocate the next block, 10.10.1.192–10.10.1.223.

Department	Subnet	Network	Valid Host Range	Broadcast	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1–10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1–10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129–10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193–10.10.1.222	10.10.1.223	30

Remaining unused range within 10.10.0.0/16: 10.10.1.224–10.10.255.255.

Final Answer: The four departments receive 254, 126, 62, and 30 usable hosts respectively.

Question 3

A smart hospital is migrating to IPv6 and has been assigned 2001:0db8:abcd:0000:0000:0000:0000:0000/64. Convert it to compressed notation, expand it, identify the /64 network prefix, and calculate the total number of addresses.

Solution:

- Given full address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64.
- Remove leading zeros in each hextet: 2001:db8:abcd:0:0:0:0:0.
- The longest consecutive sequence of zero hextets is compressed using ::.
- Compressed notation: 2001:db8:abcd::/64.
- Expanded notation: 2001:0db8:abcd:0000:0000:0000:0000:0000/64.
- The first 64 bits are the network prefix: 2001:0db8:abcd:0000::/64.
- The remaining 64 bits are available for host/interface addresses.
- Total possible addresses = 2^64 = 18,446,744,073,709,551,616.

Final Answer: Compressed address = 2001:db8:abcd::/64. Network prefix = 2001:0db8:abcd:0000::/64. Total possible addresses = 2^64.

Question 4

An ISP core router has routing table entries 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24. A packet arrives with destination IP 192.168.2.45. Determine the destination subnet, matching entry, next-hop network/interface, and default route if there is no match.

Solution:

- Destination IP = 192.168.2.45.
- With a /24 prefix, the first three octets identify the network.

- Therefore, 192.168.2.45 belongs to the 192.168.2.0/24 subnet.
- The matching routing table entry is 192.168.2.0/24.
- The router forwards the packet through the interface/next hop associated with the 192.168.2.0/24 route.
- If no listed route matches a destination, the router should use a default route, normally represented as 0.0.0.0/0, if configured.

Final Answer: Destination subnet = 192.168.2.0/24; matching entry = 192.168.2.0/24; forward using the interface/next hop configured for that route; otherwise use 0.0.0.0/0 as the default route.

Question 5

Two office departments are connected through a Layer-3 router. Administration LAN is 192.168.10.0/26 with gateway 192.168.10.1. Finance LAN is 192.168.10.64/26 with gateway 192.168.10.65. Determine network/broadcast addresses, host ranges, three suitable computer IPs in each LAN, and default gateways.

Solution:

LAN	Network	Valid Host Range	Broadcast	Default Gateway
Administration	192.168.10.0/26	192.168.10.1–192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65–192.168.10.126	192.168.10.127	192.168.10.65

Suitable Administration computer IP addresses:

- PC1: 192.168.10.2
- PC2: 192.168.10.3
- PC3: 192.168.10.4
- Default gateway: 192.168.10.1

Suitable Finance computer IP addresses:

- PC1: 192.168.10.66
- PC2: 192.168.10.67
- PC3: 192.168.10.68
- Default gateway: 192.168.10.65

Final Answer: Administration uses 192.168.10.0/26 and gateway 192.168.10.1; Finance uses 192.168.10.64/26 and gateway 192.168.10.65. The suggested PC addresses are valid host addresses in their respective LANs.

Overall Answer Summary

The five scenarios apply IPv4 subnet allocation, IPv6 addressing, routing-table matching, and Layer-3 gateway configuration. The answers are arranged as Question → Solution → Final Answer for easy study and submission.